Canadian Science Publishing | Applied Physiology, Nutrition, and Metal

Validity of a Three-Dimensional Body Scanner: Comparison Against a 4-Compartment Model and Dual Energy X-Ray Absorptiometry

Journal:	*Applied Physiology, Nutrition, and Metabolism*
Manuscript ID	apnm-2020-0744.R1
Manuscript Type:	Article
Date Submitted by the Author:	27-Oct-2020
Complete List of Authors:	Cabre, Hannah; University of North Carolina at Chapel Hill, Exercise and Sports Science Blue, Malia; University of North Carolina, Exercise and Sport Science Hirsch, Katie; University of North Carolina at Chapel Hill, Department of Exercise and Sport Science Brewer, Gabrielle; University of North Carolina at Chapel Hill, Exercise and Sport Science Gould, Lacey; University of North Carolina at Chapel Hill Nelson, Alyson; University of North Carolina at Chapel Hill, Exercise and Sport Science Smith-Ryan, Abbie; University of North Carolina Chapel Hill, Exercise and Sport Science
Novelty bullets: points that summarize the key findings in the work:	3D body composition estimates are not valid compared to the 4-compartment criterion model., 3D estimates appeared to be more valid in females, compared to males., When compared to DXA, 3D estimates were acceptable.
Keyword:	body composition, lean muscle mass < body composition, percent body fat, fat mass < body composition, fat free mass < body composition, three dimensional scanner, 3D body scanner, sex differences
Is the invited manuscript for consideration in a Special Issue? :	Not applicable (regular submission)

SCHOLARONE™ Manuscripts

© The Author(s) or their Institution(s)

Validity of a Three-Dimensional Body Scanner: Comparison Against a 4-Compartment Model and Dual Energy X-Ray Absorptiometry

Hannah E. Cabre[1,,2], Malia N. M. Blue[1,2], Katie R. Hirsch[1,2], Gabrielle J. Brewer[1], Lacey M. Gould[1], Alyson G. Nelson[1], Abbie E. Smith-Ryan[1,2]

[1]Applied Physiology Laboratory, Department of Exercise and Sport Science, University of North Carolina, 209 Fetzer Hall, Chapel Hill, NC, 27599;

[2]Human Movement Science Curriculum, Department of Allied Health Science, University of North Carolina, Chapel Hill, NC 27599

Corresponding Author
Abbie E. Smith-Ryan, PhD, FACSM, FNSCA, FISSN
Associate Professor
209 Fetzer Hall, CB#8700
University of North Carolina
Chapel Hill, NC 27599
919-962-2574
abbsmith@email.unc.edu

HEC: saylor16@live.unc.edu
MNMB: mblue@highpoint.edu
KRH: krhirsch@uams.edu
GJB: gabrielle.brewer@uconn.edu
LMG: lmgould@live.unc.edu
AGN: alyson01@live.unc.edu
ASR: abbsmith@email.unc.edu

Current Affiliations:
Malia N. M. Blue: Human Biomechanics and Physiology Lab, Department of Exercise Science, High Point University, One University Parkway, High Point, NC 27268

Katie R. Hirsch: University of Arkansas for Medical Sciences, 4301 West Markham Street, Little Rock, AR 72205-7199

Gabrielle J. Brewer: Korey Stringer Institute, Department of Kinesiology, University of Connecticut, Storrs, CT 06269

Draft

© The Author(s) or their Institution(s)

Abstract

Three-dimensional (3D) body scanner technology for body composition assessment is expanding. The aim of this study was to assess the validity of a 3D body scanner. 194 participants (43% male; Age: 23.52±5.47 yrs; BMI: 23.98±3.24 kg·m^2) were measured using 3D scanner and a 4-compartment (4C) model utilizing DXA, air displacement plethysmography, and bioelectrical impedance spectroscopy. Dependent t-tests, validity statistics including total error (TE), standard error of the estimate (SEE), constant error, and Bland-Altman analyses were utilized. Compared to 4C, 3D scanner FM [mean difference (MD; 3D- 4C): 2.66 kg±3.32 kg] and %BF (MD: 4.13%±5.36%) were significantly ($p<0.001$) over-predicted; FFM was significantly underpredicted (MD: -3.15 kg±4.75 kg; $p<0.001$). 3D demonstrated poor validity indicated by TE (%BF: 5.61%; FM: 4.50 kg; FFM: 5.69 kg). In contrast, there were no significant differences between 3D and DXA measures; 3D scanner demonstrated acceptable measurement for %BF (TE: 4.25%), FM (TE: 2.92 kg), and LM (TE: 3.86 kg). Compared to the 4C criterion, high TE values indicated 3D estimates were not valid. In contrast, 3D estimates produced acceptable measurement agreement when compared to DXA; an average overestimation of %BF by 5.31% (vs. 4C) and 4.20% (vs. DXA) may be expected.

Draft

Novelty:

- 3D body composition estimates are not valid compared to the 4-compartment criterion model.
- 3D estimates appeared to be more valid in females, compared to males.
- When compared to DXA, 3D estimates were acceptable.

Keywords: Body composition, lean mass, percent body fat, fat mass, fat-free mass, three dimensional scanner, 3D body scanner, sex differences

© The Author(s) or their Institution(s)

Introduction

Body composition measurement provides vital information regarding health status and disease risk (Pi-Sunyer 2019). Two-compartment (2C) body composition technologies, such as air displacement plethysmography (ADP) or bioelectrical impedance, are commonly used due to time and ease of measurement. 2C technology estimates two compartments of the body including fat mass (FM) and fat-free mass (FFM), while relying on assumptions for other compartments such as total body water (TBW) and bone mineral content (BMC); these assumptions often limit measurement validity of 2C technology (Forslund 1998, Okorodudu et al. 2010, Smith-Ryan et al. 2017). Three compartment technology [i.e. dual energy x-ray absorptiometry (DXA)] additionally includes measurement of bone mineral density, potentially increasing validity (Lee & Gallagher 2008). A four compartment (4C) model containing estimates of total body water (TBW) is considered the most valid criterion method, demonstrating similar validity compared to a six compartment model (Wang et al. 1998, Wang et al. 2002, Blue et al. 2018). Despite the strong validity, a 4C approach requires considerable time, equipment, and expertise. Recently, new three-dimensional (3D) body scanner technologies have gained popularity for the extensive amount of body composition measurements provided in a short amount of time.

3D scanner technology utilizes visible or infrared light to illuminate the subject within view of the scanner's cameras (Heymsfield et al. 2018). There are currently two methods by which the scanners collect data: structured light scanning or time of flight scanning. Structured light scanners calculate the depth of images of the human body by evaluating light deformation patterns. The time of flight scanners, such as Styku®, measure the time it takes for reflected photons to travel to the human then back to the image sensor (Heymsfield et al. 2018). The entire

© The Author(s) or their Institution(s)

scan of the body surface takes less than a minute; software algorithms then produce a 3D image of the body topography via meshing (Wells 2012). Anthropometric data, such as circumferences, volumes, and surface areas can be obtained from the image using landmark procedures; regression-based estimates of body composition are also provided.

Previous research examining the validity of body composition outcomes from 3D structured light scanners and time of flight scanners utilize various reference methods such as hydrostatic weighing (HW) (Wang et al. 2006, Harbin et al. 2018), ADP (Ryder and Ball 2012, Adler et al. 2017), bioelectrical impedance analysis (BIA), (Medina-Inojosa 2016, Harbin et al. 2018), and DXA (Garlie et al. 2010, Ryder and Ball 2012, Lee et al. 2015, Medina-Inojosa 2016, Tinsley et al. 2020). To our knowledge, only one previous study has validated structured light scanners and time of flight 3D technology against a 4C criterion model in healthy adults; the time of flight scanner, Styku, exhibited acceptable LOA for %BF and FM (Tinsley et al. 2020). However, the Styku measurements may not be valid against the 4C criterion due to large total error (TE) for %BF= 6.1%, FM= 4.6 kg, and FFM= 4.6 kg; structured light scanners (FIT3D®, Naked Labs®, and Size Stream®) provided more valid TE values (average %BF: 4.4%) (Tinsley et al. 2020). Styku has also previously been validated against less sophisticated criterion methods; %BF estimates in young healthy adults compared to HW were significantly lower [mean difference (MD; 3D- HW): -3.9% ± 0.7%] and were significantly lower when compared to bioelectrical impedance analysis (BIA) (MD; 3D- BIA: -2.0% ± 1.3%) (Harbin et al. 2018).

The density of LM varies across ages, sex, and obesity; accounting for LM in the validation of body composition measures is important for understanding validity (Grinker 1990). Variation in leanness has been shown to affect the accuracy of body composition measurements, particularly DXA, largely due to differences in fat and fat-free mass density and trunk estimates

© The Author(s) or their Institution(s)

(Wells and Fewtrell 2006). While previous research has validated Styku estimates of circumferential, regional, and whole-body volume estimates compared to the DXA, which is the primary method utilized in the development for the 3D prediction algorithms (Bourgeois et al. 2017), Styku estimates of body composition have yet to be validated. Additionally, analysis using the updated Styku algorithm have not been previously explored. The primary aim of this study was to assess the validity of a time of flight 3D body scanner (Styku) compared to a 4C criterion estimates of body composition including %BF, FM, and FFM. A secondary aim was to evaluate the validity against DXA, and to evaluate validity of the 3D scanner when stratified by for sex and LM. It was hypothesized that the 3D body scanner would overestimate %BF and FM, but underestimate FFM and/or LM compared to the results from the 4C model criterion, as well as for males/females, and those with high amounts of LM.

Draft

Methods

Subjects

194 males and females (43% male; Age: 23.52 ± 5.47 yrs; Weight: 69.46 ± 12.37 kg; Height: 169.92 ± 9.69 cm, BMI: 23.98 ± 3.24 $kg \cdot m^{-2}$) (table 1) participated in body composition assessments as part of three approved studies (IRB #: 19-1144; 19-0148; 17-0952). The participants' BMIs ranged from normal to obese (BMI: 18.5 – 34.85 $kg \cdot m^{-2}$). 58% of the sample was Caucasian, 16% Asian, 13% African American, and 13% Hispanic. Participants were excluded from the study that they had lost or gained five kilograms in the three months prior to enrollment, if they were taking medication that affected hydration status, had a metal implant/pacemaker, or if they had made significant changes to their dietary habits in the previous

© The Author(s) or their Institution(s)

three months. All females completed a pregnancy test once they were enrolled to confirm non-pregnant status.

Experimental Design

Data presented in this cross-sectional study were collected from a single visit. Participants completed one morning testing session and were asked to arrive euhydrated after a twelve hour fast. Participants were also asked to abstain from alcohol, tobacco, smoking, caffeine, and vigorous exercise for at least 12 hours prior to the testing day. Participants signed a written informed consent prior to all procedures. Upon arrival, height and weight were measured, and BMI was calculated. Prior to body composition measures, participants removed all metal and jewelry. Body composition was evaluated by three methods: 1) 3D body scanner, 2) a three-compartment model (DXA), and 3) 4C model criterion. All procedures were approved by the University of North Carolina's Biomedical Institutional Review Board.

Draft

Four-Compartment Body Composition Model

A 4C body composition criterion model described by Wang et al. (Wang et al. 2002) was used to determine FM, BF, and FFM (equations 1-3):

(1) $FM\ (kg) = 2.748(BV) - 0.699(TBW) + 1.129(Mo) - 2.051(BM)$

(2) $\%BF = (FM/BM) \times 100$

(3) $FFM\ (kg) = BM - FM$

where BV = body volume; TBW = total body water; Mo = total body bone mineral density; BM = body mass, measured from the ADP calibrated scale. BV was assessed with ADP, TBW was

© The Author(s) or their Institution(s)

assessed with BIS, and BM was assessed on a calibrated scale. Bone mineral content (BMC), estimated from a total body DXA (GE Lunar iDXA, GE Medical Systems Ultrasound and Primary Care Diagnostics, Madison, WI, USA) scan, was used to calculate total bone mineral density (Mo= total body BMC × 1.0436). The DXA scan was also used to measure FM and lean mass (LM; FFM - BMC). Lean mass index (LMI; $kg \cdot m^{-2}$) measured by the DXA was used to stratify LM using published cutoffs from the National Health and Nutrition Examination Survey (NHANES) based on normative values for sex, race, and age (Fan et al. 2014). According to NHANES, low LMI was defined as <50th percentile (n= 63), normal LMI was defined as 50th percentile to the 90th percentile (n=115), and high LMI was defined as >90th percentile (n=16). A trained DXA technician positioned the participants in the center of the DXA table with all limbs adjusted to fit inside the scanning area. Previous data from this lab has shown high test-retest reliability for the 4C model; FM (intraclass correlation coefficient (ICC)= 0.99, standard error of measurement (SEM)= 0.831 kg), FFM (ICC=0.99, SEM=0.99 kg), BV (ICC=0.99, SEM=0.62 L), and %BF (ICC=0.98, SEM=0.96 %) (Smith-Ryan et al. 2017).

Draft

Bioelectrical Impedance Spectroscopy

A multi-frequency bioelectrical impedance device (SFB7, ImpediMed, Queensland, Australia) was used to assess TBW for the 4C model criterion. Participants laid supine on a table with legs separated, arms spread apart from the torso, and palms of the hands flat against the table. Four electrodes were placed on the right-side of the participant's body: two electrodes were placed 5 cm apart on the wrist and hand and two electrodes were placed 5 cm apart on the ankle and foot according to manufacture guidelines. The test was completed twice with the average of both

© The Author(s) or their Institution(s)

measurements recoded for TBW. Previous data from this laboratory has shown high test-retest reliability for BIS measures (ICC= 0.99, SEM= 0.93 L) (Smith-Ryan et al. 2017).

Air Displacement Plethysmography

An air-displacement device (BodPod; COSMED USA Inc., Concord, Calif., USA) was used to assess BM and BV for the 4C model criterion. Prior to testing, the device and scale were calibrated according to manufacture guidelines. Participants were measured wearing tight fitting clothing such as a bathing suit or compression shorts and a swimming cap to cover their hair. Prior to testing, weight was measured to the nearest 0.01 kg using the device's corresponding scale (Tanita Inc., Tokyo, Japan). Participants were asked to enter the egg-shaped device, sit in an upright position, minimize movement, and breathe normally. BV was measured by a minimum of two trials within 150 mL of each other. Thoracic gas volume was estimated by the software's standard prediction equation, which has been previously compared to measured lung volume showing no significant difference between the two measures.(Mccrory et al. 1998, Demerath et al. 2002)

Draft

Three-Dimensional Body Scanner

3D body scanning device (Styku; Styku, Software Version 4.1+, Phoenix Advanced, Los Angeles, CA, USA) device was used to estimate total body FM, LM, BMC, and %fat; FFM was calculated by adding the estimated BMC with LM. Participants were asked to remove all metal and jewelry and were measured wearing tight fitting clothing such as a bathing suit or compression shorts. Female participants were asked to wear their hair in a bun. Participants stood on the center of a turn table with arms separated from the torso and hands in a fist. The turntable

© The Author(s) or their Institution(s)

measured the participant's body weight followed by a 360-degree rotation. The scan took approximately 45 seconds to complete (figure 1). After the scan, the participant's birth date, height, and origin of birth were entered into the Styku software.

Statistical Analysis

The 3D body scanner was compared to the traditional 4C criterion model for body composition outcomes including %BF, FM, FFM, and LMI using dependent samples t-test; 3D outcomes were also evaluated against DXA with the same statistical technique for %BF, FM, LM, and LMI. Validity statistics including total error (TE $=\sqrt{\sum[3D\text{-}4C \text{ or } DXA]^2/n}$), standard error of the estimate (SEE $=\sqrt{\sum[4C \text{ or } DXA\text{-}]}\bullet\sqrt{1\text{-}r^2}$), and constant error (CE= 3D- 4C or DXA) were completed to assess the validity of the 3D body scanner derived measures of %BF, FM and FFM compared to the 4C criterion and DXA. Heyward and Wagner (Heyward and Wagner 2004) standards for evaluating prediction errors were used to classify the TE and SEE for %BF, FM, and FFM for 4C and LM for DXA as ideal, excellent, very good, good, fairly good, fair, or poor. A subsequent exploratory analysis was completed removing individual outliers that were two standard deviations above or below the mean. Bland-Altman plots were also created to determine the mean difference and 95% limits of agreement (LOA) between 4C model criterion vs. 3D body scanner and DXA vs. 3D body scanner for %BF, FM, FFM, and LM. A weak Pearson correlation coefficient was defined as r = 0.0 - 0.3, a moderate correlation as r = 0.4- 0.7, and a strong correlation as r = 0.8 - 1.0 (Ratner 2009). Significance level was set at an alpha level of 0.05. The t-test and validity statistics were performed in Microsoft Excel (Microsoft Corporation, Redmond, WA, USA).

Results

© The Author(s) or their Institution(s)

Draft

Total Sample

The 3D estimates of %BF (mean ± standard deviation (SD): 25.52% ± 7.77%) were significantly greater ($p<0.001$) compared with %BF measured by 4C (mean ± SD: 21.39% ± 8.76%), but were not significantly different compared with %BF from DXA (%BF: 25.62% ± 8.21%, $p=0.74$). 3D FM measures (mean ± SD: 17.45 kg ± 5.95 kg) were also significantly greater ($p<0.001$) than the 4C (FM: 14.79 kg ± 6.91 kg), but there were no significant differences between DXA and 3D measures of FM (17.74 kg ± 6.67 kg, $p= 0.18$). 3D estimates of FFM (51.52 kg ± 11.60 kg) were significantly less ($p<0.001$) than the 4C criterion model (FFM: 54.67 kg ± 11.90 kg). There were no significant difference between 3D measures of LM (49.11 kg ± 11.24 kg) and DXA measures of LM (49.21 kg ± 11.04 kg, $p=0.71$). Individual level validity for each variable is presented using Bland-Altman Plots (Figure 2A - 2F). Constant error, TE, SEE, and Pearson's correlation coefficient of 4C model and DXA compared to the 3D body scanner outcomes for the total sample (n=194) are presented in table 2. Compared to the 4C, 3D estimates of %BF, and FFM resulted in poor measurement error; 3D FM estimates resulted in fair agreement. Compared to DXA, 3D estimates of %BF resulted in fairly good measurement error, while 3D FM resulted in good estimates; 3D LM resulted in fair estimates. With outliers that were two standard deviations above or below the mean removed [4C (n= 20) and DXA (n= 11)], the validity of the 3D scanner improved for %BF (4C TE= 4.65%; DXA TE= 3.58%), FM (4C TE= 3.43 kg; DXA TE= 2.36 kg), FFM (TE= 4.44 kg), and LM (TE=2.83 kg).

Stratified by Sex

In males, 3D measures of %BF and FM were significantly greater ($p<0.001$) compared to 4C %BF and FM while 3D FFM was significantly less ($p<0.001$) than 4C FFM (table 3). Compared

© The Author(s) or their Institution(s)

to DXA, 3D measures of %BF, FM, and LM were not significantly different compared to DXA measures (p=0.35, p= 0.11, p=0.76, respectively). Compared to the 4C criterion, 3D %BF and FFM resulted in poor measurement error; FM estimates were fair (table 4). Compared to DXA, 3D %BF, and LM were fairly good; FM estimates were good.

In females, 3D measures of %BF and FM were significantly greater (p<0.001) compared with 4C %BF and FM while 3D FFM measures were significantly less (p<0.001) compared to 4C FFM. Compared to DXA, 3D measures of %BF, FM, and LM were not significantly different (p=0.68, p= 0.83, p=0.84, respectively). 3D %BF and FFM resulted in fair measurement error compared to the 4C criterion; FM estimate was good. Compared to DXA, 3D %BF estimate resulted in fairly good measurement precision; 3D FM and LM estimates resulted in very good measurement of error.

Draft

Stratified by Lean Mass Index

In low and normal LMI, 3D measures of %BF, FFM, and FM were significantly different (p<0.01) compared with 4C. In high LMI, %BF and FM were significantly different (p<0.01) compared with 4C, but there were no significant differences for FFM. Compared to DXA, 3D measures of %BF, LM, and FM were significantly different (p<0.05) in low LMI; in normal LMI, there were only significant differences (p<0.01) for LM. There were no significant differences between 3D and DXA measures in the high LMI group. 3D %BF and FFM resulted in poor measurement error compared to the 4C criterion for all LMI categories (table 5); FM estimates were fair for all LMI categories. Compared to DXA, 3D %BF and LM estimates for low LMI resulted in poor measurement precision, while FM estimates were good. For normal

© The Author(s) or their Institution(s)

LMI, 3D estimates of %BF were good, FM estimates were very, and LM were fairly good compared to DXA. In high LMI, 3D estimates of %BF and FM were good compared to DXA; LM estimates were poor.

Discussion

Exploration of 3D body scanners for body composition assessment is rapidly expanding (Adler et al. 2017, Tinsley et al. 2020), largely due to their relatively low cost and high portability. The present study demonstrated poor validity between a 4C criterion and the 3D scanner. Specifically, FM (mean difference (MD; (3D- 4C): 2.66 kg ± 3.32 kg) and %BF (MD: 4.13% ± 5.36%) were significantly over-predicted from a 3D assessment, and FFM was underpredicted (MD: -3.15 kg ± 4.75 kg). FM estimates from 3D in females were more valid compared to males but still suboptimal in both sexes. Collectively 3D estimates compared to 4C produced unacceptable error for all body composition variables, regardless of sex or LMI. In contrast, compared to the DXA, 3D estimates were very good in both males and females; wide limits of agreement were present, suggesting some caution should be used with interpreting the agreement between these devices. 3D comparisons with DXA demonstrated better agreement for those with normal to high LM.

3D body composition estimates from the current study do not appear to be valid against a 4C model; CE, TE, and SEE values demonstrated poor to fair agreement for %BF, FM, and FFM, with significant group mean differences between methods. These results align with one recently published study validating the Styku against a 4C model in normal-weight adults who reported similar TE for %BF = 6.1%, FM= 4.6 kg, and FFM= 7.3 kg (Tinsley et al. 2020). Other 3D technologies (Fit3D, Naked, and Size Stream) have demonstrated better validity against a 4C

© The Author(s) or their Institution(s)

criterion for estimates of %BF (average TE: 4.43%), FM (average TE: 3.37 kg), and FFM (average TE: 3.37 kg) (Tinsley et al. 2020). Differences in validity between 3D technologies could result from variance in machine hardware, data collection methods, and device anthropometric algorithms. In the current study, 3D estimation of %BF showed the largest measurement error with wide LOA and high CE, suggesting there is a potential for a 4.13% overestimation of %BF in normal-weight and overweight adults. When the Styku was previously compared against other laboratory methods of underwater weighing, BIA, and skinfolds in a young, healthy population, estimates of %BF were significantly lower from Styku ($p<0.001$; LOA: -14.51%- 5.10%), with greater error in those with higher adiposity (Harbin et al. 2018). When VitusSmart XXL 3D measures of %BF were compared to ADP in normal-weight to overweight individuals, there was an average 7% over-estimation in 3D %BF estimates ($p<0.001$; LOA: 5.0- 9.1) (Adler et al. 2017). Additionally, a study comparing Hamamatsu Bodyline 3D measures of %BF to ADP in normal weight adults showed similar results to the present study with a 4% overestimation of %BF (LOA: -4.1% ± 19.6%) (Wells et al. 2006). The differences in %BF estimates observed between 3D technologies could be a result of deviations in BV measurements between devices, thereby causing proportionally pronounced differences in %BF estimates. In the present study, the large TE for 3D estimates of FM (TE: 4.50 kg) and FFM (TE: 5.69 kg) suggests 3D estimates are not comparable with 4C criterion body composition values. While the validity of 3D estimates improved upon removal of outliers, the inclusion of the outliers in the analysis reflects the true measurement error of the 3D device; discussion of the full data set is important for practical use of the device among a sample with varied anthropometric characteristics.

Draft

© The Author(s) or their Institution(s)

When compared to DXA, 3D body composition estimates from the current study appear to demonstrate more acceptable validity than when compared to the 4C model; 3D estimates of FM (TE: 2.92 kg) demonstrated good agreement; LM (TE: 3.86 kg) and %BF (TE: 4.25%) resulted in fair to fairly good agreement, respectively. These results appear to be similar to a previous study comparing FIT3D Proscanner to DXA in normal-weight to obese adults for %BF (TE: 3.06%) and FM (TE: 2.36 kg) (Ng et al. 2016). Furthermore, when 3D stereovision body imaging was compared to DXA in healthy adults, higher acceptable group-level SEE for %BF= 4.22% (LOA: -0.11 ± 1.97%) and FM= 3.39 kg (LOA: -0.06 ± 0.87 kg) were reported (Lee et al. 2015). Other 3D technologies appear to have similar validity compared to the present investigation when DXA was used as a reference method for %BF. To our current knowledge, there are no comparisons for 3D estimates of lean mass. The present study showed fair estimations of LM compared to DXA (TE: 3.86 kg). With existing technology and algorithms, 3D scanners appear to provide acceptable body composition estimates in comparison to DXA. However, LOA were quite wide for 3D-DXA comparisons; likely exceeding changes that may be expected to occur as a result of lifestyle or nutritional interventions. Caution should be taken when using the Styku to monitor individual changes in body composition.

Previous studies have reported varied agreement for body composition measurement when accounting for physiological sex (Zillikens et al. 2008). When compared to the 4C criterion, Styku body composition estimates from the present study reported better measurement agreement for females vs. males. Specifically, FM estimates in females were considered acceptable. While males and females in the present study were normal-weight, females had higher %BF and FM. Styku landmark algorithms estimating body surface and body composition measurements have been validated primarily in overweight/obese individuals. Male participants

© The Author(s) or their Institution(s)

in a previous validation study had a higher %BF than the present study (Bourgeois et al. 2017); therefore, the higher measurement error observed for males could be attributed to the origin of the algorithm data points, and could be influenced by larger LM. Compared to DXA, 3D measures in females demonstrated very good estimates for FM (SEE= 2.48 kg) and LM (SEE= 3.06 kg), compared to fairly good estimates in males FM (SEE=3.36 kg) and LM (SEE= 4.19 kg). 3D %BF estimates for both sexes were fairly good with a SEE of 4.14% and 4.25% for females and males, respectively. Similarly, a study validating a Textile/Clothing Technology Corp [TC^2] 3D body scanner estimates %BF against DXA in 97 young, healthy adult males reported similar error (SEE= 4.59%) (Ryder and Ball 2012). In contrast, poor validity was observed (SEE=6.61%) for 3D estimates of %BF in 70 healthy normal-weight to obese Caucasian and Hispanic females compared to DXA (Mitchell et al. 2011). To our knowledge, this is the first study to compare 3D validity between sexes, suggesting 3D estimates as a potentially viable estimate of FM and LM in females and greater potential for error in males.

The present sample was stratified by LMI to identify the Styku's ability to estimate body composition in populations with disparate LM. Due to standard assumptions of body compartments, accuracy of body composition measurement often varies across leanness. Compared to 4C criterion, 3D body composition estimates in the current study were not valid in any LMI category. The poor measurement agreement across LMI categories may be attributed to hydration status not being accounted for with 3D measurements. Compared to DXA, 3D estimates demonstrated similar validity regardless of LMI resulting in very good estimates of FM and good estimates of %BF. Measurements of LM were acceptable in normal LMI (SEE= 3.30 kg), but unacceptable/poor in both low and high LMI (SEE= 3.91 kg and 4.87 kg, respectively).

© The Author(s) or their Institution(s)

The present findings suggest 3D estimates are a potentially practical measurement of FM and %BF regardless of LM, when compared to DXA, but is a poor estimator of LM.

The 4C criterion model has been shown to be the most accurate method for assessing body composition due to measuring more compartments of the body; the potential cumulative error associated with the use of multiple techniques is outweighed by the increased accuracy of measuring, and not assuming, each compartment (Wilson et al. 2012). While the 4C criterion is considered the gold standard, its use is limited by the amount of time and cost incurred due to the variety of equipment needed especially compared to a 40-second scan. The inclusion of DXA in the current study as a comparison method provided additional validity for settings that cannot implement the 4C criterion. Body mass measurements could have had a minor influence on error as body mass was obtained from software internal to the respective devices (i.e. ADP vs. Styku); the difference between weight utilized in the 4C/DXA and the weight measured by the Styku was not significant (mean difference (3D – 4C or DXA); -0.49 ± 0.12 kg; p= 0.70). Furthermore, while the current study was considered to be relatively diverse (58% Caucasian, 16% Asian, 13% African American, and 13% Hispanic) variations in body composition previously identified in racial and ethnic groups (Heymsfield et al. 2016) were not explored in the present investigation due to insufficient sample size. Future research would benefit from evaluating the validity of Styku in all racial/ethnic groups.

Conclusion

The TE and SEE results in the present study indicate that 3D body composition estimates were not valid compared to the 4C criterion, regardless of sex or LMI, but estimates were acceptable when compared to DXA. 3D estimates appeared to be more acceptable in females,

© The Author(s) or their Institution(s)

compared to males; 3D FM estimates in both sexes were valid compared to DXA. In subjects with high or low amounts of LM, estimates were not valid against the 4C; in contrast, 3D estimates of FM were acceptable across all categories of LMI. 3D body scanners offer a quick tool to estimate body composition and anthropometrics; the ease of use and portability may be beneficial for field-based and large group assessments, such as outpatient settings. While 3D Styku body composition estimates provide valid utility against DXA, for research purposes or scenarios requiring more sensitive determination, Styku estimates are not warranted.

Acknowledgements: There was no direct funding to disclose for this manuscript. The Styku device was given in kind to the University laboratory.

Conflict of interest: The Authors declare that there is no conflict of interest.

Draft

© The Author(s) or their Institution(s)

References

Adler C., Steinbrecher A., Jaeschke L., Mähler A., Boschmann M., Jeran, S., Pischon, T. 2017. Validity and reliability of total body volume and relative body fat mass from a 3-dimensional photonic body surface scanner. PLoS ONE, 12(7): e0180201. doi: 10.1371/journal.pone.0180201.

Blue M.N.M., Hirsch K.R., Trexler E.T., Smith-Ryan A.E. 2018. Validity of the 4-compartment model using dual energy x-ray absorptiometry-derived body volume in overweight individuals. Appl. Physiol. Nutr. Metab., 3(7):742-746. doi: 10.1139/apnm-2017-0804.

Bourgeois B., Ng B.K., Latimer D., Stannard C.R., Romeo L., Li X., Shepherd J.A., Heymsfield S.B. 2017. Clinically applicable optical imaging technology for body size and shape analysis: Comparison of systems differing in design. Eur. J. Clin. Nutr., 71(11):1329-1335. doi: 10.1038/ejcn.2017.142.

Demerath E.W., Guo S.S., Chumlea W.C., Towne B., Roche A.F., Siervogel R.M. 2002. Comparison of percent body fat estimates using air displacement plethysmography and hydrodensitometry in adults and children. Int. J. of Obes., 26(3):389-97. doi: 10.1038/sj.ijo.0801898.

Fan B., Shepherd J., Levine M., Steinberg D., Wacker W., Barden H., Ergun D., Wu X.P. 2014. National Health and Nutrition Examination Survey Whole-Body Dual-Energy X-Ray Absorptiometry Reference Data for GE Lunar Systems. J. Clin. Dens., 14(3):344-377. doi: 10.1016/j.jocd.2013.08.019.

Forslund, A. H. 1998. Evaluation of modified multicompartment models to calculate body composition in healthy males. Appl. Radiat. Isot., 49(5-6):599-60. doi: 10.1016/S0969-8043(97)00079-1.

Draft

© The Author(s) or their Institution(s)

Garlie T.N., Obusek J.P., Corner B.D., Zambraski E.J. 2010. Comparison of body fat estimates using 3D digital laser scans, direct manual anthropometry, and DXA in men. Am. J. Hum. Biol., 22(5):695-701. doi: 10.1002/ajhb.21069.

Grinker, J. A. 1990. Body Composition Measurement: Accuracy, Validity, and Comparability - Body Composition and Physical Performance - NCBI Bookshelf, Institute of Medicine (US) Committee on Military Nutrition Research. Available at: https://www.ncbi.nlm.nih.gov/books/NBK235949/

Harbin, M. M., Kasak A., Ostrem J., Dengel, D.R. 2018. Validation of a three-dimensional body scanner for body composition measures. Eur. J. Clin. Nutr., 72: 1191-1194. doi: 10.1038/s41430-017-0046-1.

Heymsfield, S. B., Peterson C.M., Thomas D.M., Heo M., Schuna Jr. J.M., 2016. Why are there race/ethnic differences in adult body mass index-adiposity relationships? A quantitative critical review. Obes. Rev. 17(3):262-275. doi: 10.1111/obr.12358.

Heymsfield S. B., Bourgeois B., Ng B.K., Sommer M. J., Li X., Sheperd J.A. 2018. Digital anthropometry: A critical review. Eur. J. Clin. Nutr., 72:680-687 doi: 10.1038/s41430-018-0145-7.

Heyward V. H. and Wagner D. R. 2004. Applied body composition assessment. 2nd Edition. Human Kinetics.

Medina-Inojosa J., Somers V.K., Ngwa T., Hinshaw L., Lopez-Jimenez F. 2016. Reliability of a 3D Body Scanner for Anthropometric Measurements of Central Obesity. *Obesity: Open Access*., 2(3).doi: 10.16966/2380-5528.122.

Lee J.J., Freeland-Graves J.H., Pepper M.R., Stanforth P.R., Xu B. 2015. Prediction of Android and Gynoid Body Adiposity via a Three-dimensional Stereovision Body Imaging System and

Draft

© The Author(s) or their Institution(s)

Dual-Energy X-ray Absorptiometry. J. Am. Coll. Nutr., 34(5):367-77. doi: 10.1080/07315724.2014.966396.

Lee, S. Y. and Gallagher, D. 2008. Assessment methods in human body composition. Curr Opin Clin. Nutr. Metab. Care., 11(5):566-72. doi: 10.1097/MCO.0b013e32830b5f23.

McCrory M.A., Molé P.A., Gomez T.D., Dewey K.G., Bernauer E.M. 1998. Body composition by air-displacement plethysmography by using predicted and measured thoracic gas volumes. J. Appl. Physiol., 84(4):1475-9. doi: 10.1152/jappl.1998.84.4.1475.

Pepper M.R., Freeland-Graves J.H., Yu W., Stanforth P.R., Xu B. 2010. Evaluation of a Rotary Laser Body Scanner for Body Volume and Fat Assessment. J. Test. Eval., 39(1):1-6. doi: 10.1520/jte102871.

Ng B. K., Hinton B.J., Fan B., Kanaya A.M. 2016. Clinical anthropometrics and body composition from 3D whole-body surface scans. Eur. J. Clin. Nutr., 70: 1265–1270 doi: 10.1038/ejcn.2016.109.

Okorodudu D.O., Jumean M.F., Montori V.M., Romero-Corral A., Somers V.K., Erwin P.J., Lopez-Jimenez F. 2010. Diagnostic performance of body mass index to identify obesity as defined by body adiposity: A systematic review and meta-analysis. Int. J. Obes. (Lond)., 34(5):791-9. doi: 10.1038/ijo.2010.5.

Pi-Sunyer X. 2019. Changes in body composition and metabolic disease risk. Eur J Clin Nutr., 73(2):231-235. doi: 10.1038/s41430-018-0320-x.

Ratner B. 2009. The correlation coefficient: Its values range between 1/1, or do they. Target Meas. Anal. Mark., 17: 139–142. doi: 10.1057/jt.2009.5.

Ryder J. R. and Ball S. D. 2012. Three-dimensional body scanning as a novel technique for body composition assessment: A preliminary investigation. J. Ex. Phys. Online., 15(1):1-14.

Draft

© The Author(s) or their Institution(s)

Smith-Ryan A.E., Mock M.G., Ryan E.D., Gerstner G.R., Trexler E.T., Hirsch K.R. 2017. Validity and reliability of a 4-compartment body composition model using dual energy x-ray absorptiometry-derived body volume. Clin. Nutr.,36(3):825-830. doi:10.1016/j.clnu.2016.05.006

Tinsley GM, Moore ML, Benavides ML, Dellinger JR, Adamson BT. 2020. 3-Dimensional optical scanning for body composition assessment: A 4-component model comparison of four commercially available scanners. Clin. Nutr., 39(10):3160-3167. doi:10.1016/j.clnu.2020.02.008

Wang J. Gallagher D., Thornton J., Yu W., Horlick, M., Xavier Pi-Sunyer F. 2006. Validation of a 3-dimensional photonic scanner for the measurement of body volumes, dimensions, and percentage body fat. Am. J. Clin. Nutrt., 83(4):809-8-16. doi: 10.1093/ajcn/83.4.809.

Wang Z.M., Deurenberg P., Guo S.S., Pietrobelli A., Wang J. Pierson Jr. R.N., Heymsfield S.B. 1998). Six-compartment body composition model: inter-method comparisons of total body fat measurement. Int. J. Obes. Relat. Metab. Disord.,22(4):329-337. doi:10.1038/sj.ijo.0800590

Wang Z. M., Pi-Sunyer F.X., Kotler D.P., Wielopolski L. Withers R. Pierson R.N. Heymsfield S. B. 2002. Multicomponent methods: evaluation of new and traditional soft tissue mineral models by in vivo neutron activation analysis. Am. J. Clin. Nutr., 76(5):968-974. doi:10.1093/ajcn/76.5.968

Wells J. C. K. 2012. Three-dimensional (3-D) photonic scanning: A new approach to anthropometry. Handbook of Anthropometry. Handbook of Anthropometry. Springer, New York, NY. doi: 10.1007/978-1-4419-1788-1_11.

Wells J.C.K., Douros I., Fuller N.J., Elia M., Dekker L 2000. Assessment of body volume using three-dimensional photonic scanning. Ann N Y Acad Sci. 904:247-254. doi:10.1111/j.1749-6632.2000.tb06460.x

Wells J.C.K., Fewtrell M.S. 2006. Measuring body composition. Arch Dis Child., 91(7):612-617. doi:10.1136/adc.2005.085522

Draft

© The Author(s) or their Institution(s)

Wilson J.P., Mulligan K., Fan B., Sherman J.L., Murphy E.J., Tai V.W., Powers C.L., Marquez L., et al. 2019.Dual-energy X-ray absorptiometry-based body volume measurement for 4-compartment body composition. Am J Clin Nutr., 95(1):25-31. doi:10.3945/ajcn.111.019273
Zillikens M.C., Yazdanpanah M., Pardo L.M., Rivadeneira F. Aulchenko Y.S., Oostra B. A., Uitterlinden A.G., Pols H.A.P, et al. 2008. Sex-specific genetic effects influence variation in body composition. Diabetologia., 51(12):2233-2241. doi:10.1007/s00125-008-1163-0

Draft

© The Author(s) or their Institution(s)

Tables

Table 1. Descriptive characteristics by sex presented as mean ± standard deviation.

Table 2. Validity statistics comparing the 4-comparment (4C) criterion and DXA to a 3-dimentional (3D) body scanner measures of percent body fat (%BF), fat mass (FM), fat free mass (FFM), and lean mass (LM) in the total sample (n= 194).

Table 3. Mean ± standard deviation of 3-dimentional (3D) body scanner, 4-comparment (4C) criterion, and DXA body composition measures of percent body fat (%BF), fat mass (FM), fat free mass (FFM), and lean mass (LM) in males vs. females.

Table 4. Validity statistics comparing the 4-comparment (4C) criterion and DXA with a 3-dimentional (3D) body scanner measure of percent body fat (%BF), fat mass (FM), and lean mass (LM) in males (n=83) and females (n=111).

Table 5. Validity statistics comparing the 4-comparment (4C) criterion and DXA with a 3-dimentional (3D) body scanner body measures of percent body fat (%BF), fat mass (FM), and lean mass (LM) in individuals with a low lean mass index (LMI) (n= 63), normal lean mass index (n= 115), and high lean mass index (n= 16).

Draft

© The Author(s) or their Institution(s)

Figures

Figure 1. Configuration of the three-dimensional body scanner (Styku®; Styku Software Version 4.1+, Los Angeles, CA, USA).

Figure 2A-2F. Bland-Altman analyses for 3D measures of percent body fat (%BF) 95% limits of agreement [LOA; (3D – 4C)]= –6.39 to 14.64 %; fat mass (FM; LOA = –4.47 to 9.79 kg), and fat free mass (LM; LOA= –14.9 to 3.78 1kg) compared to four compartment and percent body fat (%BF) 95% limits of agreement [LOA; (3D – DXA)]= –8.46 to 8.25 %; fat mass (FM; LOA = –5.99 to 5.42 kg), and lean mass (LM; LOA= –7.68 to 7.48 kg) compared to the dual energy x-ray absorptiometry.

Draft

© The Author(s) or their Institution(s)

Table 1. Descriptive characteristics by sex presented as mean ± standard deviation.

	Males (n= 83)	**Females (n= 111)**
Age (yrs)	23.8 ± 6.1	23.2 ± 4.9
Weight (kg)*	76.9 ± 11.6	63.8 ± 9.6
Height (cm)*	177.4 ± 7.8	164.2 ± 6.5
BMI (kg/m^2)	24.4 ± 3.3	23.6 ± 3.1
4C %BF*	15.0 ± 7.3	26.1 ± 6.4

* Indicates significant differences between sex ($p<0.05$).

Draft

© The Author(s) or their Institution(s)

Table 2. Validity statistics comparing the 4-comparment (4C) criterion and DXA to a 3-dimentional (3D) body scanner measures of percent body fat (%BF), fat mass (FM), fat free mass (FFM), and lean mass (LM) in the total sample (n= 194).

	3D vs 4C			3D vs DXA		
	%BF	FM (kg)	FFM (kg)	%BF	FM (kg)	LM (kg)
CE	-4.13	-2.66	3.15	0.10	0.28	0.10
TE	5.61[e]	4.50[e]	5.69[e]	4.25[c]	2.92[b]	3.86[d]
SEE	5.31[e]	3.64[d]	4.76[e]	4.20[c]	2.91[b]	3.77[d]
R	0.80*	0.85*	0.92*	0.86	0.90	0.90
R^2	0.63	0.72	0.84	0.74	0.81	0.88

Note: Estimates were derived as 3D – 4C; 3D- DXA. *R*, Pearson's correlation coefficient; R^2, coefficient of determination; CE, constant error; TE, total error; SEE, standard error of the estimate; *, $p<0.001$. Classifications according to Heyward and Wagner 2004: [a], Very good; [b], Good; [c], Fairly good ; [d], Fair; [e], poor.

Draft

© The Author(s) or their Institution(s)

Table 3. Mean ± standard deviation of 3-dimentional (3D) body scanner, 4-comparment (4C) criterion, and DXA body composition measures of percent body fat (%BF), fat mass (FM), fat free mass (FFM), and lean mass (LM) in males vs. females.

	Males (n= 83)			**Females (n= 111)**		
	3D	**4C**	**DXA**	**3D**	**4C**	**DXA**
%BF*	18.90 ± 5.56	15.07 ± 7.37	19.35 ± 6.14	30.47 ± 5.00	26.12 ± 6.45	30.31 ± 6.18
FM (kg)*	14.60 ± 5.18	11.82 ± 6.54	15.20 ± 6.07	19.59 ± 5.61	17.01 ± 6.34	19.63 ± 6.49
FFM (kg)*	61.78 ± 9.29	65.17 ± 9.74	-	43.86 ± 5.77	46.82 ± 5.71	-
LM (kg)	59.04 ± 9.03	-	59.19 ± 8.78	41.69 ± 5.56	-	41.75 ± 5.01

* Indicates significant differences between 3D and 4C measures ($p<0.001$).

Draft

© The Author(s) or their Institution(s)

Table 4. Validity statistics comparing the 4-comparment (4C) criterion and DXA with a 3-dimentional (3D) body scanner measure of percent body fat (%BF), fat mass (FM), and lean mass (LM) in males (n=83) and females (n=111).

	Males					
	3D vs 4C			3D vs DXA		
	%BF	FM (kg)	FFM (kg)	%BF	FM (kg)	LM (kg)
CE	-3.83	-2.78	-3.39	0.46	0.59	0.15
TE	7.10[e]	5.23[e]	6.53[e]	4.40[c]	3.40[b]	4.36[c]
SEE	5.90[e]	4.44[c]	5.54[e]	4.25[c]	3.36[b]	4.19[c]
R	0.60*	0.73*	0.82*	0.72	0.83	0.88
R^2	0.36	0.54	0.68	0.52	0.69	0.77
	Females					
	3D vs 4C			3D vs DXA		
	%BF	FM (kg)	FFM (kg)	%BF	FM (kg)	LM (kg)
CE	-4.35	-2.57	2.97	-0.16	0.05	0.07
TE	6.49[e]	3.86[d]	4.96[e]	4.15[c]	2.50[a]	3.43[c]
SEE	4.79[d]	2.89[b]	3.77[d]	4.14[c]	2.48[a]	3.06[a]
R	0.67*	0.89*	0.75*	0.74	0.92	0.79
R^2	0.45	0.79	0.56	0.55	0.85	0.63

Note: Estimates were derived as 3D- 4C; 3D – DXA. *R*, Pearson's correlation coefficient; R^2, coefficient of determination; CE, constant error; TE, total error; SEE, standard error of the estimate; *, $p<0.001$. Classifications according to Heyward and Wagner 2004: [a], Very good; [b], Good; [c], Fairly good ; [d], Fair; [e], poor.

Draft

© The Author(s) or their Institution(s)

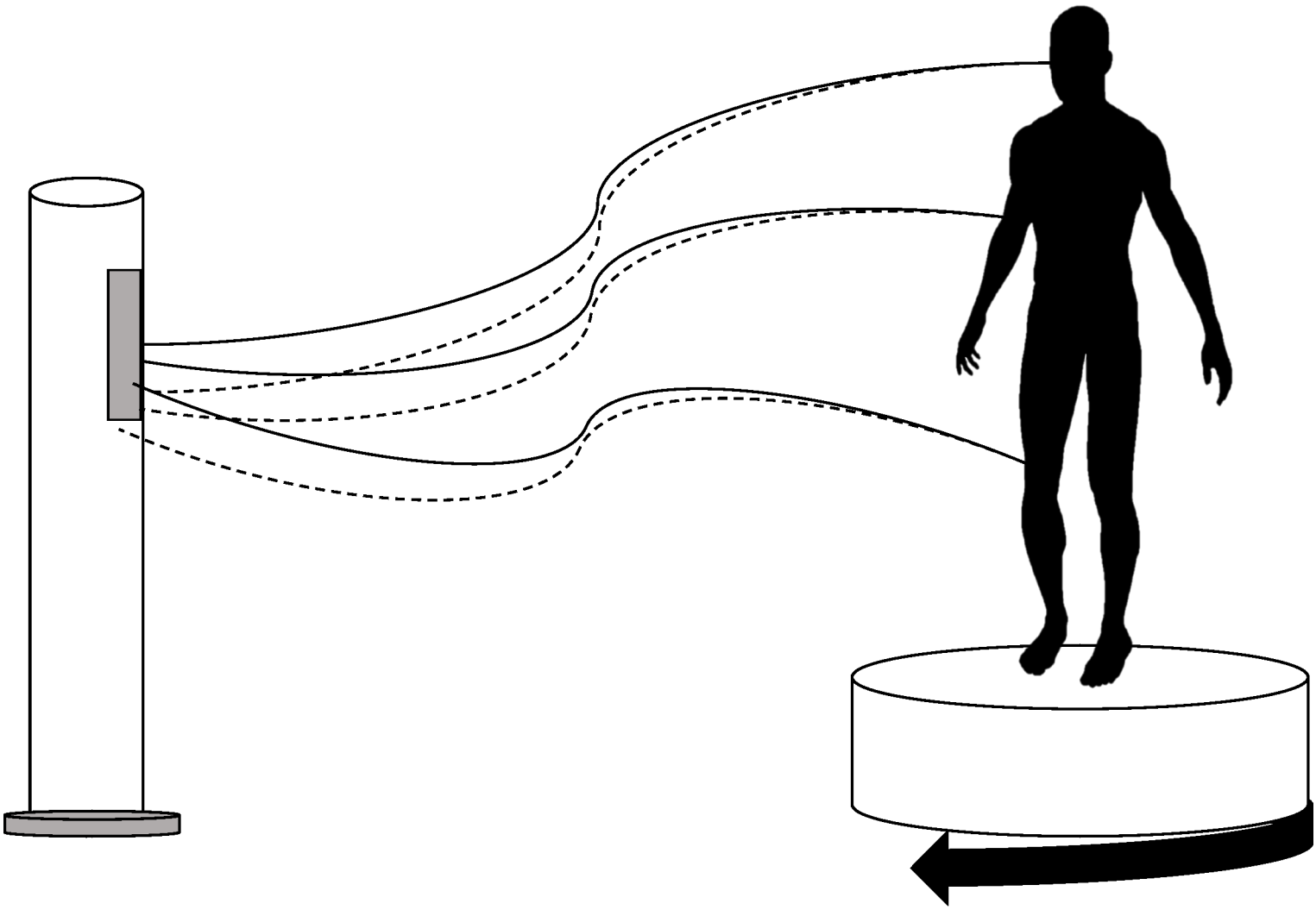
Table 5. Validity statistics comparing the 4-comparment (4C) criterion and DXA with a 3-dimentional (3D) body scanner body measures of percent body fat (%BF), fat mass (FM), and lean mass (LM) in individuals with a low lean mass index (LMI) (n= 63), normal lean mass index (n= 115), and high lean mass index (n= 16).

	3D vs 4C								
	Low LMI			Normal LMI			High LMI		
	%BF	FM(kg)	FFM(kg)	%BF	FM(kg)	FFM(kg)	%BF	FM(kg)	FFM(kg)
CE	-2.26	-1.58	1.67	-4.96	-3.10	4.03	-4.30	-3.16	2.96
TE	6.40[e]	4.07[d]	5.24[e]	6.91[e]	4.54[e]	5.93[e]	6.57[e]	5.14[e]	5.75[e]
SEE	5.89[e]	3.72[d]	5.01[e]	4.80[d]	3.32[c]	4.34[d]	5.07[e]	3.30[d]	4.82[e]
R	0.76[*]	0.74[*]	0.88[*]	0.82[*]	0.88[*]	0.91[*]	0.86[*]	0.95[*]	0.93[*]
R^2	0.58	0.55	0.77	0.67	0.77	0.83	0.73	0.90	0.86
	3D vs DXA								
	Low LMI			Normal LMI			High LMI		
	%BF	FM(kg)	LM(kg)	%BF	FM(kg)	LM(kg)	%BF	FM(kg)	LM(kg)
CE	1.65	1.08	-1.01	-0.65	-0.10	0.85	-0.40	0.15	-0.40
TE	5.12[e]	3.20[b]	4.08[e]	3.73[b]	2.62[a]	3.48[c]	3.93[b]	3.48[c]	4.92[e]
SEE	4.64[d]	2.88[b]	3.91[d]	3.69[b]	2.60[a]	3.30[c]	3.98[b]	2.79[a]	4.87[e]
R	0.81[*]	0.80[*]	0.90[+]	0.89	0.93	0.94[*]	0.90	0.96	0.92
R^2	0.66	0.64	0.82	0.79	0.86	0.89	0.82	0.92	0.85

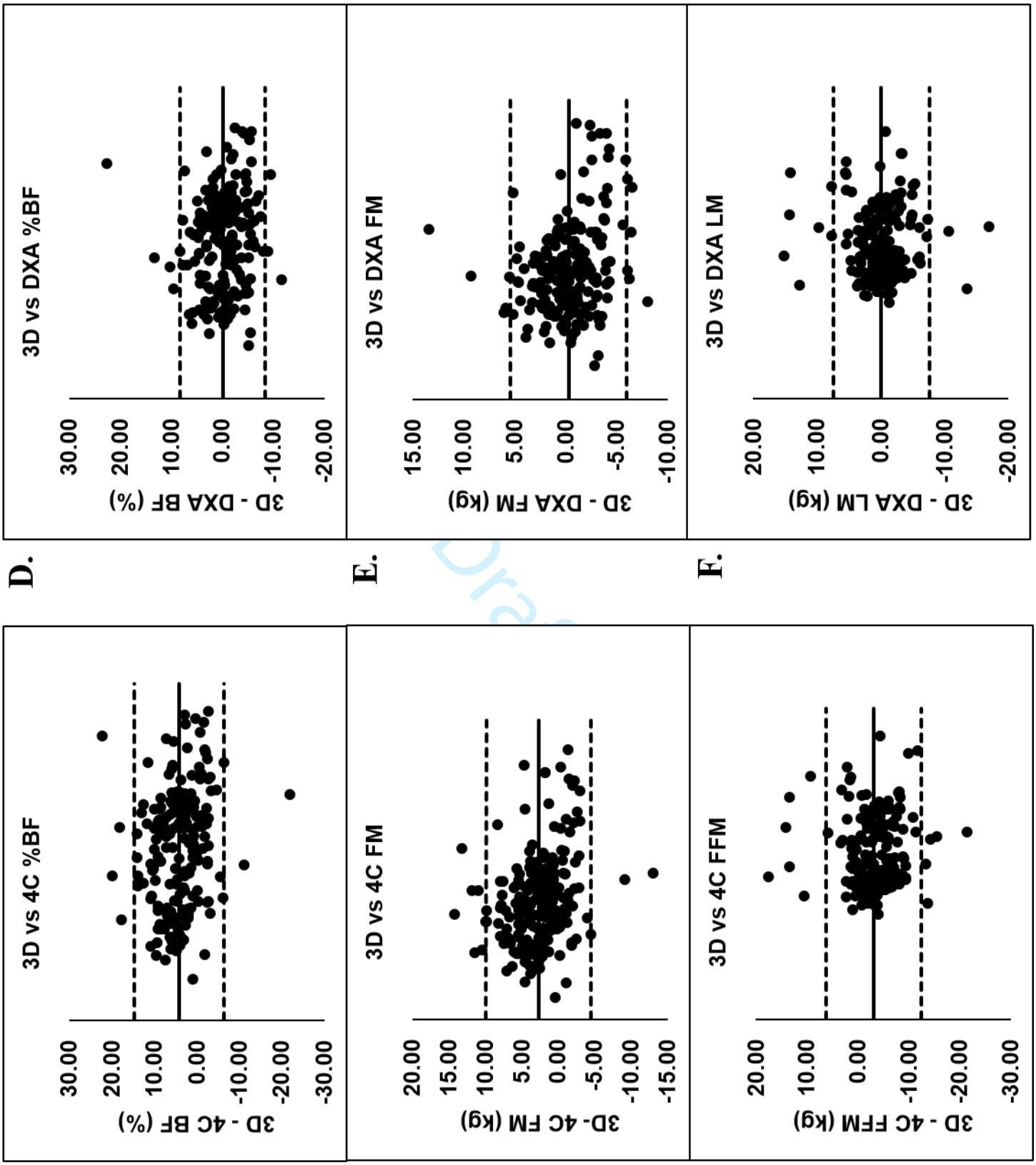
Note: Estimates were derived as 3D – 4C; 3D - DXA. *R*, Pearson's correlation coefficient; R^2, coefficient of determination; CE, constant error; TE, total error; SEE, standard error of the estimate; [*], $p<0.001$. Classifications according to Heyward and Wagner 2004: [a], Good; [b], Fairly good; [c], Fair; [d], Poor.

Draft

© The Author(s) or their Institution(s)



© The Author(s) or their Institution(s)



© The Author(s) or their Institution(s)